

modelling also uses deep neural networks but with competitive and unsupervised learning.

The key factors driving deep learning adoption are:

- The requirement for sophisticated computer vision applications in various fields like genetics, robotics, etc
- The requirement for complex predictive deep learning models across various industries
- Advances in speech and natural language processing
- Autonomous vehicle control
- The capability to process data gathered from connected devices
- Next generation (intelligent) industrial revolution

Deep learning requires large amounts of data (labelled, unlabelled or semi-labelled) for training, testing and to come up with accurate predictions. For example, driverless car development requires millions of images and thousands of hours of video.

Deep learning requires huge computing power. High-performance GPUs have a parallel architecture that is efficient for deep learning. When combined with clusters or cloud computing, this enables development teams to reduce training time for a deep learning network from weeks to hours or less. For example, the availability of data sets on deep learning network architectures helps in multi-object detection and tracking in real-time with more accuracy and high speed.

Much more research is going into many areas of deep learning such as self-driving cars, genetics, medical diagnosis, connected devices, etc.

The characteristics of deep learning are:

- Requires large amounts of data and compute power
- Automatic feature extraction (feature learning)
- Auto-correction of hyper parameters (backpropagation)
- Trainable with custom data (transfer learning)

- Works well with both structured and unstructured data
- Hierarchical feature learning

Types of deep learning

We will primarily focus on the discriminative models and will touch upon generative models briefly.

Supervised models: Like supervised machine learning models, supervised deep learning models use labelled/semi-labelled data to train. These fall into the discriminative models category. Given below are various models.

- **Multi-layer perceptron (or classical neural networks):** This is a shallow learning model with very few hidden layers. This model is applicable to deep learning problems like regression, classification, etc. These layers are fully connected across entire networks.
- **Convolutional neural networks:** These are deep neural networks (CNN) and consist of an input layer, multiple hidden layers and an output layer. The layers are convolutional layers, pooling layers and fully connected layers. This model can handle complex data processing as well as unstructured data like images, voice, etc.
- **Recurrent neural networks:** A recurrent neural network (RNN) is

designed to accept a series of inputs of variable sizes. Here, a series means a set of related inputs that influence/relate to one another. RNNs operate on a hidden state with shared weights for predictions. This model is used for sentiment analysis, NLP, etc.

Unsupervised models:

Unsupervised models are used for sample generation, market segmentation, and to cluster/segment large data sets without labels. These fall in the deep generative models category. The various models are given below.

- **Self-organising maps (SOM):** A self-organising map is a model trained using unsupervised learning to produce low dimensional input data. It is used for dimensionality reduction.
- **Boltzmann machines (BM):** Boltzmann machines are non-deterministic (or random) generative deep learning models with only two types of nodes — hidden and visible. Nodes can capture all the parameters, patterns and correlations between the input data. These models are used to generate synthetic data sets close to the target ones, and feature identification on large data sets.

Another variation of the Boltzmann machine is a restricted Boltzmann machine (RBM) used for dimensionality

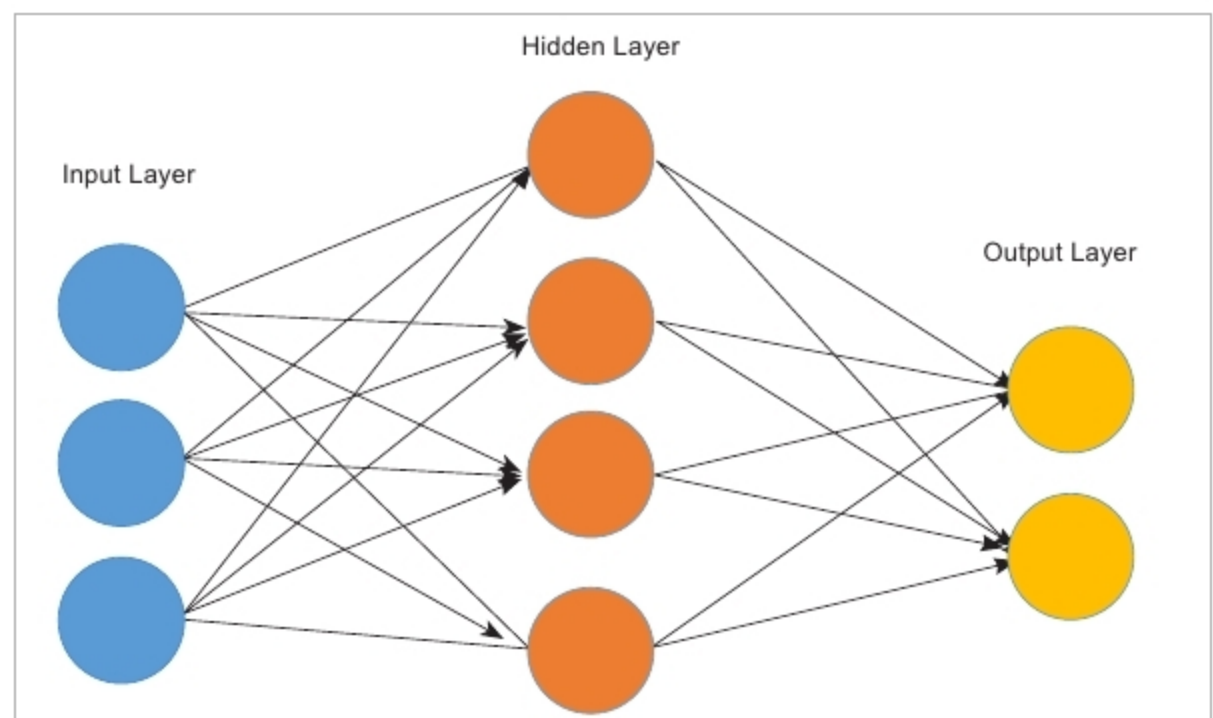


Figure 1: Convolutional neural network architecture